

Software Engineering Lab 3

Component Modelling & Architecture Selection

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Problem Statement #34: Digital Gift Card & Loyalty Point Exchange

Architecture Selection

For the **Digital Gift Card & Loyalty Point Exchange System**, we chose **Layered Architecture**. The system works like a digital wallet where users can collect loyalty points from different participating brands, convert those points into universal exchange credits using dynamic rates, and use the credits to purchase or redeem digital gift cards.

The system also includes an anti-fraud mechanism called **voucher locking**. This ensures that once a voucher has been successfully redeemed, it cannot be used again or accidentally redeemed through another conflicting transaction.

1. Architectural Choice

Selected Architectural Style: Layered Architecture

We selected Layered Architecture because it divides the system into separate layers, with each layer handling a specific type of responsibility.

- **Presentation Layer:** This is the part of the system that users and merchant partners interact with. It handles things such as login and sessions, viewing loyalty-point balances, requesting point exchanges, and redeeming gift cards.
- **Business Logic Layer:** This layer contains the main rules and processes of the system. It is responsible for checking loyalty points, calculating exchange rates, converting points into universal credits, generating and redeeming gift cards, performing anti-fraud checks, locking vouchers, and processing transactions.
- **Data Layer:** This layer manages the information stored by the system. It keeps records of loyalty-point balances, exchange transactions, gift cards and vouchers, redemption status, audit logs, and ledger entries.

By separating these responsibilities, the system becomes easier to understand and manage. The presentation layer communicates with the business logic through defined interfaces, while the business logic accesses stored information through the data layer.

2. Reasons for Choosing Layered Architecture

Reason 1 – Clear Separation of Different Operations

The platform has to handle several different activities, such as collecting loyalty points, checking point balances, calculating exchange rates, converting points into universal credits, generating gift cards, and processing redemptions.

If all of these operations were placed in one large module, the system would become difficult to manage. Layered Architecture keeps these responsibilities separated, allowing each part of the system to focus on what it does best and communicate with the other parts through clear interfaces.

Reason 2 – Easier Maintenance and Future Expansion

The system may grow over time by adding new merchant partners, loyalty programs, gift-card providers, or new rules for calculating exchange rates.

With Layered Architecture, these changes can usually be made within the relevant layer without having to redesign the entire system. For example, adding a new gift-card provider would mainly affect the business or integration-related components rather than the entire application.

This makes the system more flexible and easier to expand as the digital wallet platform grows.

3. Security Advantage

Layered Architecture also helps improve the security of the system. User requests do not directly modify important records in the database. Instead, requests pass through the business logic layer, where they can be checked and validated before any changes are made.

For example, before completing an exchange or redemption, the system can verify the user's session, check whether the required loyalty points or credits are available, apply the correct exchange rate, and perform the necessary anti-fraud checks.

Some important security benefits include:

- **Voucher locking:** Prevents the same digital gift card from being redeemed more than once, including situations where two transactions happen at nearly the same time.
- **Point and balance validation:** Helps prevent users from converting points or credits that they do not actually have.
- **Transaction and audit logs:** Keep a record of exchanges and redemptions, making it easier to track suspicious or incorrect activity.
- **Separation of presentation and data access:** Prevents the user interface from directly changing important database records and ensures that changes go through the appropriate business rules.

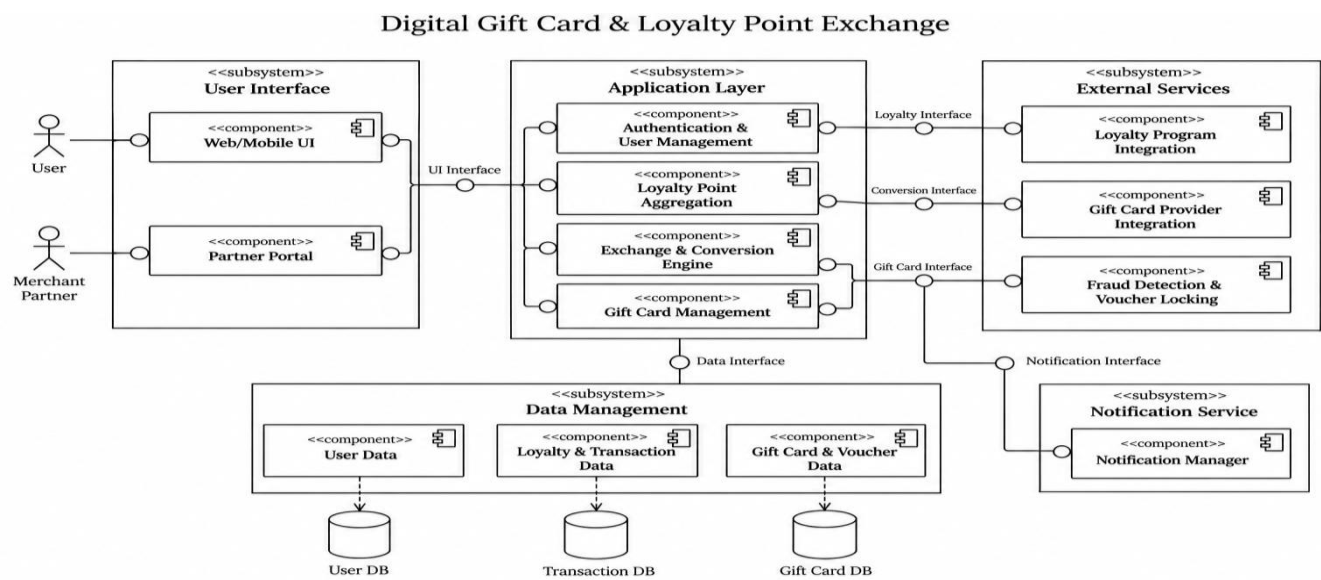
4. Performance Benefit

Layered Architecture can improve the system's operational efficiency by assigning high-frequency tasks to dedicated services. The Loyalty Point Manager can quickly validate balances, while the Exchange Rate Service can calculate the conversion value without involving presentation logic. The Gift Card Service and Voucher Lock / Anti-Fraud component can focus on secure voucher generation and redemption.

This separation reduces unnecessary processing between unrelated operations and makes it easier to optimize individual components. For example, frequently accessed loyalty balances can be retrieved efficiently from the data layer, while transaction logging can be handled independently. The result is a responsive exchange and redemption workflow, especially when multiple account holders and merchant partners use the platform.

5. Component Diagram

The component diagram below represents the proposed Layered Architecture. Account Holders and Merchant Partners interact with the Presentation Layer. Requests then move to business components such as the Loyalty Point Manager, Exchange Rate Service, Gift Card Service, Voucher Lock / Anti-Fraud, and Ledger & Transaction Service. Validated transactions are persisted through the Data Layer.



6. Component Responsibilities

Layer	Component	Primary Responsibility
Presentation	Account Holder UI	View loyalty balances, exchange points, generate/redeem gift cards.
Presentation	Merchant Partner UI	Support merchant-side interaction with loyalty and gift-card transactions.
Presentation	Authentication & Session	Authenticate users and maintain authorized sessions.
Business Logic	Loyalty Point Manager	Validate and aggregate loyalty points from participating brands.

Business Logic	Exchange Rate Service	Apply the current dynamic rate and calculate universal exchange credits.
Business Logic	Gift Card Service	Generate and process digital gift-card vouchers.
Business Logic	Voucher Lock / Anti-Fraud	Lock voucher state and prevent duplicate or invalid redemption.
Business Logic	Ledger & Transaction Service	Record atomic exchange and redemption transactions.
Data	Loyalty Point Ledger	Persist point balances and point-related transactions.
Data	Gift Card Database	Persist voucher codes, status, value, and redemption state.
Data	Transaction / Audit Logs	Maintain transaction history and security/audit records.

Conclusion

Layered Architecture is a suitable choice for the Digital Gift Card & Loyalty Point Exchange System because it provides a clean separation between presentation, business processing, and data management. It supports the core requirements of aggregating loyalty points, applying dynamic exchange rates, converting points into universal exchange credits, and securely redeeming digital gift cards. The architecture also strengthens security by placing validation and anti-fraud controls in the business layer before persistent data is updated. By isolating these responsibilities, the system becomes easier to develop, test, maintain, secure, and extend with additional merchant partners and loyalty programs.